VENKATA NARESH BODDEPALLI

Plant Breeder | Legume Specialist | Genomics, Phenomics & AI-Assisted Breeding

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PROFESSIONAL SUMMARY

Results-driven plant breeder with ~12 years of experience in legume crop improvement, integrating classical breeding with GWAS, genomic prediction, and AI-assisted high-throughput phenotyping. Proven track record advancing breeding pipelines from crossing through variety release across South Asia and the USA, with hands-on expertise in mungbean, urdbean, vegetable soybean, and cowpea. Demonstrated ability to lead cross-functional teams, manage multi-location international trials, and deliver disease-resistant, higher-yielding commercial lines. Seeking an industry plant breeder role to apply cutting-edge genomic and phenomic tools to accelerate genetic gain and cultivar development.

CORE COMPETENCIES

Breeding Pipeline Management | GWAS & Genomic Prediction | Molecular Marker Development | AI-Assisted High-Throughput Phenotyping | Germplasm Characterization & Gene Bank Management | Hybrid/Variety Development | Mutation Breeding | Multi-Location & International Trials | RNA-seq & Bioinformatics | Crop Protection Integration | Cross-Functional Team Leadership | Grant & Report Writing

PROFESSIONAL EXPERIENCE

Plant Breeding Graduate Research Assistant — May 2022 – Aug 2025

Iowa State University – Dept. of Agronomy | Ames, IA

- Introduced innovative crossing protocols that improved hybridization success from 25% to 65% and optimized early-generation advancement, shortening breeding timelines by 1–2 years.
- Contributed to soybean breeding program through controlled crosses, rouging, single-plant selection, and harvest coordination to advance high-performing commercial lines.
- Designed and implemented AI-enabled pod and seed imaging systems for automated phenotyping, reducing manual scoring time by 60%.
- Conducted GWAS and genomic prediction on 372 mungbean genotypes (~30K SNPs), integrating image-based phenomics with genomic data for quantitative trait mapping.
- Developed and validated molecular markers via RNA-seq, gene cloning, BiFC assays, and complementation studies in Arabidopsis.
- Established mutation breeding platform using physical mutagens (gamma rays, fast neutrons) to generate novel genetic variation; screened M2 populations for yield and plant architecture traits.
- FAA-licensed drone pilot: supported field-scale imaging campaigns and spatial data analysis across multi-acre nurseries.
- Standardized mungbean/urdbean BBCH growth stage charts for Midwest conditions — now used as a reference protocol by collaborators.
- Contributed to grant proposals (DARPA, USAID-NIFA, COALESCE, AIIRA) and organized AI-in-agriculture outreach for growers, researchers, and graduate students.

Scientific Officer – Legume Breeding & Seed Production (South/Central Asia) — Feb 2014 – Dec 2022

World Vegetable Center (AVRDC), ICRISAT Campus | Hyderabad, India

- Managed full breeding and seed production pipeline for 5 legume crops (mungbean, urdbean, vegetable soybean, cowpea, yard-long bean), delivering disease-resistant, higher-yielding lines to national partners across South and Central Asia.
- Designed trial layouts, field books, and data-quality standards enabling phenotypic data from multiple locations to be merged and used for selection decisions.
- Established multi-environment screening partnerships with state universities to identify viral and fungal disease-resistant lines and strengthen gene bank collections.

- Managed seed increase, purity checks, and MTA/SMTA compliance as regional gene bank coordinator, ensuring material advanced without genetic contamination.
- Introduced digital data capture and standardized phenotyping workflows adopted by 10 partner institutions, improving data timeliness and cross-site comparability.
- Led, hired, and managed a team of 10 staff (4 on-campus, 6 field sites) across multiple concurrent breeding projects.
- Trained 50+ researchers from 10 countries in mungbean breeding and KDDart database management; trained 100+ farmers on good agricultural practices.
- Co-authored successful grant proposals securing ~USD 1 million across two phases of ACIAR (Australian Centre for International Agricultural Research) funding for mungbean improvement in South/Southeast Asia.
- Contributed to RKVY-funded project proposals (Government of India) securing ~INR 4 crore (~USD 480K) for mungbean and vegetable cowpea productivity improvement in Odisha and Karnataka.
- Prepared budgets, progress reports, and donor updates for multi-partner ACIAR, RKVY, and USAID-funded projects; authored research articles, seed production manuals, and project proposals.

Graduate Teaching Assistant – AGRON 5110 Crop Improvement **Fall 2024 – Present**

Iowa State University | Ames, IA

- Delivered guest lectures on GWAS, comparative genomics, and genomic prediction; graded assignments and facilitated student discussions.

Teaching Associate – Department of Horticulture **Aug 2013 – Feb 2014**

Dr Rama Naidu Vignana Jyothi Institute for Rural Development | India

- Taught 7 credit hours: breeding & seed production of horticultural crops, protected cultivation/forestry, and communication skills; mentored 12 diploma students through PAMP practicum.

Junior Research Fellow (M.Sc. Research) **Jul 2011 – Jul 2013**

Chandra Shekhar Azad University of Agriculture & Technology | Kanpur, India

- Conducted eggplant and garden pea crossing trials; screened brinjal hybrids for Cercospora leaf spot resistance and field peas for powdery mildew resistance.

EDUCATION

Ph.D. in Plant Breeding – Phenomics & Genomics-Assisted Breeding **Expected Dec 2026**

Iowa State University, Ames, IA

Dissertation: Genomics and AI-Assisted Phenomics for Mungbean Crop Improvement | Advisor: Dr. Arti Singh

M.Sc. in Horticulture – Vegetable Breeding **Jul 2013**

Chandra Shekhar Azad University of Agriculture & Technology, Kanpur, India

Thesis: Heterosis and Combining Ability for Quantitative Traits and *Cercospora* Leaf Spot Resistance in Brinjal

B.Sc. (Hons.) in Horticulture – Protected Cultivation of Vegetable Crops **Jun 2011**

Dr YSR Horticultural University, West Godavari, India

TECHNICAL SKILLS

Breeding Tools	Hybridization, pedigree & backcross breeding, bulk method, GWAS, MAS, genomic prediction, mutation breeding (gamma rays, fast neutrons), winter nursery management, product advancement & commercialization
Phenotyping	High-throughput phenotyping (HTP), UAV/drone-based field imaging (FAA-licensed pilot), AI-enabled pod/seed imaging pipelines, digital phenotyping, precision agriculture
Molecular / -Omics	SSR/SNP marker development, primer design, gene cloning, RNA-seq, BiFC assays, Arabidopsis transformation, comparative mapping, legumeinfo.org
Bioinformatics & Data	GWAS pipelines, genomic prediction, SNP genotyping, R (statistical analysis), machine learning for phenotyping, KDDart database, data-driven decision-making
Field Operations	Planter & combine operation, multi-location trial coordination, RCBD/alpha lattice/augmented design, crop protection integration

Management	Cross-functional team leadership, project budgeting, donor reporting, grant writing, stakeholder training & engagement

KEY RESEARCH PROJECTS

- DARPA Bio-shield – Defense Advanced Research Projects Agency (Research Assistant, ISU)
- COALESCE: Context Aware Learning for Sustainable Cyber-Agricultural Systems (Research Assistant, ISU)
- AIIRA: AI Institute for Resilient Agriculture (Research Assistant, ISU)
- USAID / NIFA – National Institute of Food and Agriculture (Research Assistant, ISU)
- International Mungbean Improvement Network (IMIN) – ACIAR, Australia (Scientific Officer, WorldVeg)
- Improved Mungbean Harvesting & Seed Production for Bangladesh, Myanmar & Pakistan – ACIAR funded (Scientific Officer, WorldVeg)
- Mungbean & Urdbean Productivity Improvement in Odisha & Karnataka – RKVY Govt. funded (Scientific Officer, WorldVeg)
- USAID Agricultural Innovation Project – AIP (Scientific Officer, WorldVeg)

PEER-REVIEWED PUBLICATIONS (18)

First Author

1. **Boddepalli V.N.,** (2026). Image-based high-throughput phenotyping enables genetic analyses of pod morphological traits in mung bean (Vigna radiata (L.) R. Wilczek). *G3: Genes|Genomes|Genetics.* doi:10.1101/2025.11.06.687031 ***[Conditionally Accepted]***
2. **Boddepalli V.N.,** (2026). Integrative genomics and deep learning-based phenotyping reveal the genetic architecture of seed traits in mungbean (Vigna radiata L.). *The Plant Genome.* Manuscript ID: TPG-2026-03-0073 ***[Under Review]***
3. **Boddepalli V.N.,** Singh A. (2025). Defining the Growth Stages of Mungbean (Vigna radiata L.) using the BBCH Scale. *Agronomy Journal.* Manuscript ID: AJ-2025-12-1229-OA. doi:10.1101/2025.11.09.687504 ***[Under Review]***
4. **Boddepalli V.N.,** Dubey A.K., Dabbas M.R. (2016). Studies of standard heterosis for quantitative traits in eggplant (Solanum melongena L.). *Int. J. Agri. Sci.* 12(1):38-41. doi:10.15740/has/ijas/12.1/38-41
5. **Boddepalli V.N.,** Dubey A., Tiwari P., Shukla I. (2015). Combining ability studies for yield and its attributing traits in brinjal (Solanum melongena L.). *Current Advances in Agricultural Sciences* 7(2):117. doi:10.5958/2394-4471.2015.00029.5
6. **Boddepalli V.N.,** Dubey A.K., Tiwari P.K., Dabbas M.R. (2014). Line x Tester analysis for yield components and Cercospora leaf spot resistance in brinjal (Solanum melongena L.). *E.J. of Plant Breeding* 5(2):230-235
7. **Boddepalli V.N.,** Dubey A.K., Tiwari P.K., Dabbas M.R., Avani P. (2013). Heterosis studies for yield components and Cercospora leaf spot resistance in brinjal (Solanum melongena L.). *Progressive Research* 8(Special):410-412

Co-Author

8. Lin Y.-P., ..., **Boddepalli V.N.,** et al. (2025). The demographic history, genomic variation, and transcontinental genotype-phenotype-environment map of mungbean. *Nature Genetics.* doi:10.1101/2025.10.16.682767 ***[Under Review]***
9. Rairdin A., de Azevedo Peixoto L., **Boddepalli V.N.,** Farmer A.D., Dutta S., Cannon S.B., Singh A. (2026). Genetic Architecture of Seed Protein in Mungbean: A Multi-Environment GWAS and Genomic Prediction Study. *The Plant Genome.* ***[Under Review]***
10. Shen Y., Ayanlade T.T., **Boddepalli V.N.,** et al. (2025). WeedNet: A foundation model-based global-to-local AI approach for real-time weed species identification and classification. *Nature Communications.* doi:10.48550/ARXIV.2505.18930 ***[Under Review]***
11. Joshi S., Salvo J.D., Shen Y., Hadadi M., **Boddepalli V.N.,** Jubery Z., Sarkar S., Singh A., Ganapathysubramanian B., Singh A.K., Krishnamurthy A. (2026). Neural 3D reconstruction and immersive VR visualization of row crops across phenological growth stages. *Smart Agricultural Technology* 102024. doi:10.1016/j.atech.2026.102024 ***[Published]***
12. Nair R.M., Chaudhari S., Devi N., Shivanna A., Gowda A., **Boddepalli V.N.,** et al. (2024). Genetics, genomics, and breeding of black gram (Vigna mungo (L.) Hepper). *Frontiers in Plant Science* 14:1273363. doi:10.3389/fpls.2023.1273363
13. Nair R.M., **Boddepalli V.N.,** Yan M.-R., Kumar V., Gill B., Pan R.S., Wang C., Hartman G.L., Silva e Souza R., Somta P. (2023). Global Status of Vegetable Soybean. *Plants* 12(3):609. doi:10.3390/plants12030609
14. Pandey A.K., Basandrai A.K., Basandrai D., **Boddepalli V.N.,** Rathore A., Adapala G., Nair R.M. (2021). Field-Relevant New Sources of Resistance to Anthracnose Caused by Colletotrichum truncatum in a Mungbean Mini-Core Collection. *Plant Disease.* doi:10.1094/pdis-12-20-2722-re
15. Nair R.M., Giri R.R., **Boddepalli V.N.,** Prasad K.V.S.V., Devulapalli R., Blummel M. (2019). Variation in grain yield, fodder quality and animal intake parameters in dual purpose mungbean and vegetable soybean in semi-arid tropical India. *Legume Research* 44:207-214. doi:10.18805/LR-4078
16. Nair R.M., Leelapriya T., Dhilip K.S., **Boddepalli V.N.,** Ledesma D.R. (2018). Beneficial effects of Extremely Low Frequency (ELF) Sinusoidal Magnetic Field exposure on mineral and protein content of mungbean seeds and sprouts. *Indian Journal of Agricultural Research* 52:126-132

17. War A.R., Murugesan S., **Boddepalli V.N.,** Srinivasan R., Nair R.M. (2017). Mechanism of Resistance in Mungbean (Vigna radiata (L.) R. Wilczek var. radiata) to bruchids, Callosobruchus spp. (Coleoptera: Bruchidae). *Frontiers in Plant Science* 8:1031. doi:10.3389/fpls.2017.01031
18. Nair R.M., Gotz M., Winter S., Giri R.R., **Boddepalli V.N.,** Sirari A., et al. (2017). Identification of mungbean lines with tolerance or resistance to yellow mosaic in fields in India where different begomovirus species and Bemisia tabaci cryptic species predominate. *European Journal of Plant Pathology* 149:349-365

CONFERENCE PAPERS (7)

1. **Boddepalli V.N.,** War A.R., Pandey A.K., Shivanna A., Gowda A., Sunil, Nair R.M. (2021). Development of black gram varieties with improved protein quality, and resistance to bruchids and mungbean yellow mosaic disease. *National Web Conference on Sustaining Pulse Production for Self-sufficiency and Nutritional Security,* ICAR-IIPR, Kanpur, India. pp.97
2. Gowda A., Kumar D., Wankhede D., Pradhan H., **Boddepalli V.N.,** Archak S., Singh N., Singh K., Chaudhari S., Nair R.M. (2021). Identification of sources of resistance against Colletotrichum lindemuthianum in black gram. *National Web Conference on Sustaining Pulse Production,* ICAR-IIPR, Kanpur, India. pp.316
3. Pandey A.K., Yee M., Win M.M., Lwin H.M.M., **Boddepalli V.N.,** Gowda A., Chaudhari S., Nair R.M. (2021). Identification of sources of resistance against dry root rot (Macrophomina phaseolina) in mungbean mini-core collection. *National Web Conference on Sustaining Pulse Production,* ICAR-IIPR, Kanpur, India. pp.330
4. Nair R.M., **Boddepalli V.N.,** War A.R., Pandey A.K. (2019). Improving black gram (Vigna mungo (L.) Hepper) against biotic stresses. *Legumes for Human and Planet Health,* Poznan, Poland. pp.36
5. **Boddepalli V.N.,** Bejjarapu K.S., Pandey A.K., Nair R.M. (2019). Identification of potential sources of resistance in mungbean against anthracnose (Colletotrichum truncatum). *XIX International Plant Protection Congress (IPPC2019),* Hyderabad, India. pp.326
6. Nair R.M., Giri R.R., **Boddepalli V.N.,** Sharada P., Prasad K.V.S.V., Reddy Y.R., Devulapalli R., Blummel M. (2015). Variation in mungbean for grain yield, haulm yield and forage quality. *9th Biennial Conference of the Animal Nutrition Association,* Guwahati, India. p.32
7. Nair R.M., Giri R.R., **Boddepalli V.N.,** Sharada P., Prasad K.V.S.V., Reddy Y.R., Devulapalli R., Blummel M. (2015). Variation in vegetable soybean for fresh pod and haulm yield and forage quality. *9th Biennial Conference of the Animal Nutrition Association,* Guwahati, India. p.33

CONFERENCES ATTENDED

- Plant and Animal Genome Conference, San Diego, CA – Poster Presentation [Jan 2026]
- 2025 Loomis Symposium, ISU, Ames, IA – Student Speaker [May 2025]
- 13th Raymond F. Baker Plant Breeding Symposium, ISU, Ames, IA – Poster Presentation [Mar 2025]
- 12th R.F. Baker Plant Breeding Symposium, ISU, Ames, IA – Poster Presentation [Mar 2025]
- 8th International Plant Phenotyping Symposium (IPPS8), Lincoln, NE – Poster Presentation [Oct 2024]
- 6th International Workshop on Machine Learning for Cyber-Agricultural Systems, Lincoln, NE [Oct 2024]
- National Association of Plant Breeders (NAPB), St. Louis, IL – Poster Presentation [Jul 2024]
- 11th R.F. Baker Plant Breeding Symposium, ISU, Ames, IA – Poster Presentation [Mar 2024]
- 10th R.F. Baker Plant Breeding Symposium, ISU, Ames, IA – Poster Presentation [Mar 2023]
- National Web Conference on Sustaining Pulse Production for Self-sufficiency, India – Poster [Feb 2021]
- XIX International Plant Protection Congress (IPPC2019), Hyderabad, India – Poster [Nov 2019]
- 9th Biennial Conference of the Animal Nutrition Association, Guwahati, India – Poster [Jan 2015]
- State-level Seminar on Sustainable Agriculture & Food Processing, ICRISAT, India – Poster [Jan 2014]

HONORS & AWARDS

- NAPB Borlaug Scholar (2026) – $1,750
- Bayer Graduate Student Travel Award (2025) – $750
- GDM Travel Grant (2025) – $1,000
- NAPB Travel Grant (2025) – $1,000
- Professional Advancement Grant (2025) – $250
- Outstanding Performance Awards (x2) – World Vegetable Center
- Recognized as "Mungbean Messenger" by the World Vegetable Center for outstanding contributions to legume breeding in South Asia.
- Best Poster Presentation – State-level Seminar on Sustainable Agriculture & Food Processing, ICRISAT, Hyderabad, January 2014.

- Book Prize Award – 1st place in Olericulture Division, M.Sc. program, CSAU&T, February 2014.
- National Eligibility Test (NET) – Qualified for Lectureship/Assistant Professorship, ICAR Agricultural Scientists Recruitment Board, June 2014.

LEADERSHIP & SERVICE

- President & Co-chair, 13th Raymond F. Baker Plant Breeding Symposium, ISU (Jul 2025 – Jun 2026) – Led 7-person graduate student committee for symposium planning and execution.
- Agronomy Department Senator & Election Committee Member, GPSS, ISU (Aug 2024 – May 2025) – Represented agronomy students and liaised with CALS Dean on student issues.
- Chair, Food & Venue Committee, 11th R.F. Baker Plant Breeding Symposium, ISU (Aug 2023 – Mar 2024).
- Member, Speaker Committee, 10th R.F. Baker Plant Breeding Symposium, ISU (Jul 2022 – Mar 2023).
- Lead Trainer, KDDart Database Management, International Mungbean Improvement Network, WorldVeg (Jan 2016 – Dec 2021) – Organized hands-on workshops in Myanmar and Bangladesh.

PROFESSIONAL AFFILIATIONS

- National Association of Plant Breeders (NAPB), Member (2022–2025)
- American Society of Agronomy (ASA), Member (2022–2025)
- Crop Science Society of America (CSSA), Member (2022–2025)
- Soil Science Society of America (SSSA), Member (2022–2025)
- Society of Scientific Development in Agriculture & Technology, Lifetime Member